

Airbnb Bookings Predictions

SVM and Neural Network Model Performance Analysis

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Problem Statement

A new competitor in the private residential market is interested in the Airbnb market in Cape Town and would like an analysis on the key predictors of demand of/ availability of Airbnbs. Demand in this case is proxied by the target variable, `fully_booked_30`, which is a binary variable that populates 1 if the Airbnb is fully booked in the next 30 days and 0 if not. This is done by using and comparing classification models:

- SVM with a linear kernel
- SVM with a radial kernel
- Feed-forward Neural Network

Exploratory Analysis

The data is highly imbalanced in favour of the negative class. Since the modelling aims to predict availability, this will be treated as the positive class in modelling- this is the target class.

The data used was listings data containing key attributes across Airbnb IDs.

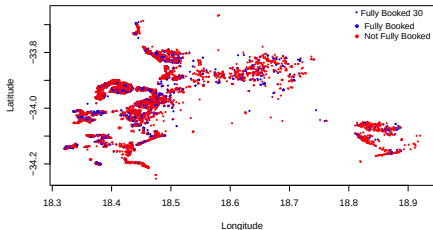


Figure 1: Spatial Plot of BNB Listings, Fully Booked 30 Label Distribution

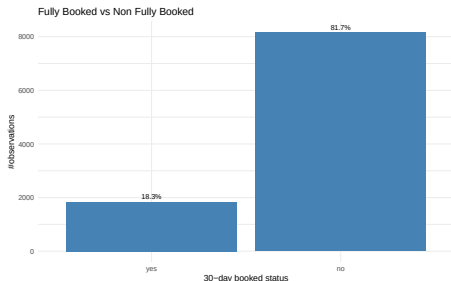


Figure 2: Spatial Plot of BNB Listings, Fully Booked 30 Label Distribution

Feature Distributions

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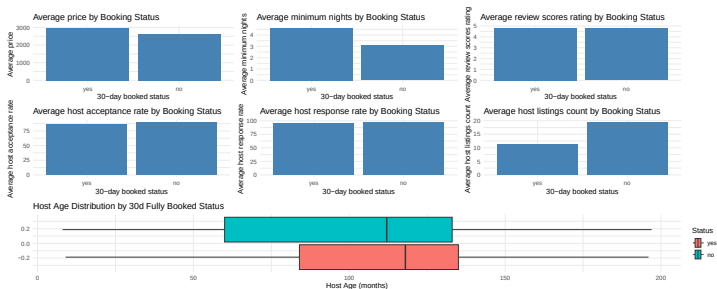


Figure 3: Feature Averages and Distributions Grouped by Target

Linear Kernel SVM

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Table 1: Linear Kernal SVM Best Model Summary

Metric	Value
model_name	Linear Kernal SVM
best_cost	0.15
nr.support_vectors	3807
best_cross_validation_error	0.18
best_training_accuracy	0.82

Model Training Linear Kernel SVM

Linear Kernel SVM Gridsearch Summary

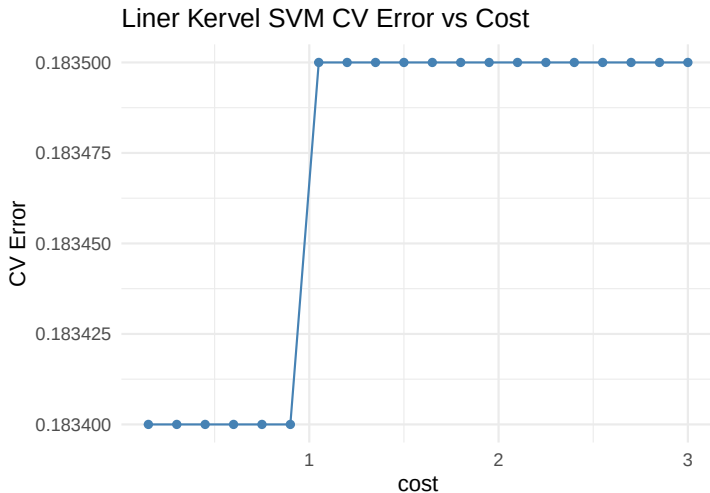


Figure 4: Linear Kernal SVM CV Error vs Cost

Non-Linear SVM Tuning Summary

Cross-validation was used to train a radial-kernel SVM on the data. This kernel has an additional similarity parameter which controls the rate of similarity between points based on their distance; thus, this needs to be tuned during the cross-validation process.

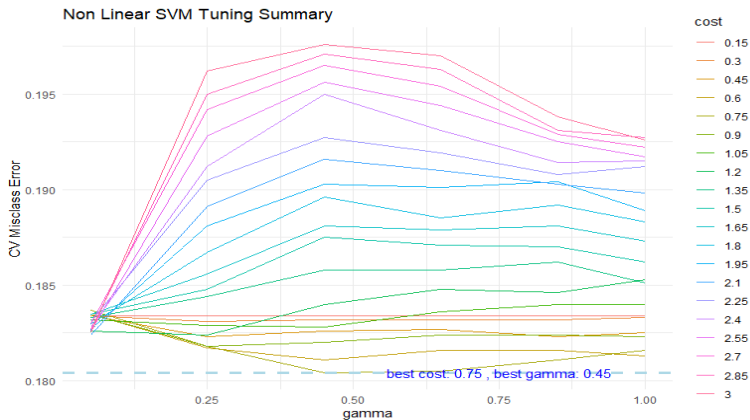
Metric	Value
nr.support_vectors	7421.00
best_cost	0.75
best_gamma	0.45
best_cross_validation_error	0.18
best_training_accuracy	0.85

Non Linear Kernel SVM CV Error vs Cost

Model Training Linear Kernel SVM

Gridsearch Training Summary

Across parameter combinations, the model starts to overfit as the gamma parameter increases past 0.50, as well as some higher values of gamma, past its optimal value.



Model Training Feed Forward Neural Network

Grid Search Tuning Summary

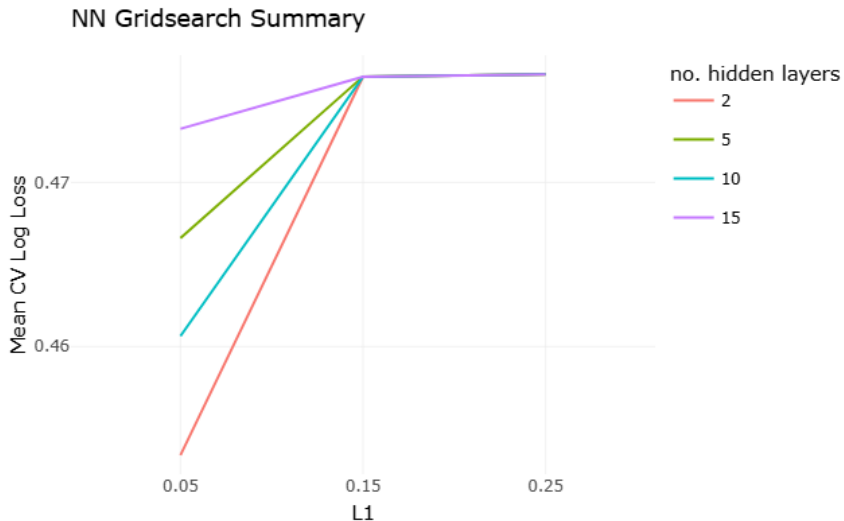


Figure 6: Neural Network Gridsearch summary

Best Model Summary

Table 3: Neural Net Best Model Metrics

Metric	Value
best L1	0.05
best nr. hidden layer nodes	2.00
best CV Accuracy	0.82
best CV Logloss	0.45

Cross Model Comparison

Table 4: Cross Model CV Metrics

Model	Specificity	Sensitivity	F1	Bal..Accu- racy
Linear	1.00	0.00	NA	0.50
Kernal SVM				
Radial	1.00	0.20	0.33	0.60
Kernel SVM				
Feed	0.73	0.51	0.65	0.62
Forward NN				

